PingSheng Kao

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SUMMARY

ML Data Scientist with 5+ years of experience improving multimodal, LLM, agentic, and production-vision systems through evaluation. Builds scalable judge workflows, user simulations, expert-aligned benchmarks, behavioral quality signals, and model feedback loops that turn failures into actionable product improvements.

EXPERIENCE

ML Data Scientist - Apple July 2020 - Present
Santa Clara, California

- Built and scaled LLM-as-judge and agent-as-judge evaluation on an existing internal batch platform, auditing long multimodal conversations for factual consistency and tool-call behavior, debugging throughput and rate-limit issues, and surfacing evaluator outputs in dashboards.
- Built a simulated-user agent, targeted scenario generators, and user-centered metrics to stress-test an open-ended multimodal assistant, probing visual-memory and sycophancy failures and estimating rates of intent misalignment and false inferred-state updates at scale.
- Built expert-aligned evaluation for a multimodal health-related AI product: an auditable, reference-based multi-stage LLM evaluator, gold-set curation and labeling workflows, and DSPy-style prompt iteration that reached 90% alignment with experts on a 40-example gold set.
- Produced model-release quality reports pairing scorecard metrics with regression attribution, embedding-cluster diagnosis, and online validation of offline-mined failures, turning model behavior into decision-ready feedback for technical leadership.
- Wrote SQL in an evaluation pipeline querying AWS Athena to calculate model-quality metrics and export results to dashboards.
- Developed embedding-driven failure discovery for Vision Pro hand tracking using similarity search, clustering, OOD, and position-level analysis; contributed to a quality scorecard with final missing-pinch rate below 2% and false positives below 5 per hour.

RESEARCH CONTRIBUTION

GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks 2025
Experiment contributor (credited as Sam Kao) - accepted to ECCV - arXiv:2512.24592

- Contributed experiments using LLMs and VLMs to generate and verify grounded failure hypotheses for fine-grained, interpretable error-slice discovery.
- Trained and fine-tuned vision models on VLM-discovered failure slices and benchmarked targeted repair against alternative slice-discovery approaches.

ADDITIONAL EXPERIENCE

Algorithm and Architecture Intern - Deep Learning Compression - Ambarella June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

EDUCATION

M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor 2018 - 2020
B.S. in Electronic Engineering - National Taiwan University 2013 - 2017
GPA: 3.9/4.0

SKILLS

Programming: Python, SQL, C++, JavaScript
LLM & Agent Evaluation: LLM-as-Judge, Agent-as-Judge, User Simulation, Tool-Call and Factual-Consistency Evaluation, Human Evaluation
ML Systems & Data: Internal Batch Evaluation, AWS Athena, PyTorch, DSPy, Experiment Design, Statistical Analysis, Tableau
Quality Methods: Benchmark and Metric Design, Failure Analysis, Embeddings, Clustering, OOD Analysis, VLM/MLLM Evaluation